STENet: A Spatial Selection and Temporal Evolution Network for Change Detection in Remote Sensing Images

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***Abstract*— Accompanied by the booming development of remote sensing (RS) imaging techniques, change detection (CD) has emerged as a conspicuous focal point in the realm of geoscience. Traditionally, extensive research has predominantly centered on extracting semantic features from individual images yet neglecting the interinput correlations latent in bitemporal imagery. This oversight gives rise to the occurrence of pseudovariant regions and the blurring of detection boundaries. To address these challenges, we propose a spatial selection and temporal evolution network, named STENet, which aims to unravel semantic correlations between bitemporal images from both spatial and temporal perspectives. Specifically, a dual pathway is crafted. The first one explores precisely the spatial localization of changes in bitemporal pairs, while the other complements the local details by generating a pseudovideo input via specific data augmentation. To boost the precision (Pre) of localizing sparsely changed targets, we further present a dynamic selective attention (DSA) mechanism, which strives for more focus on the positive regions while holding a mild computational demand. Moreover, by leveraging data augmentation to derive the temporal evolution of a set of progressively changed images, we then exploit a 3-D convolution-based encoder to mine the potential details therein, endeavoring to a refinement of target boundaries. Ultimately, STENet enforces the fusion of multiscale spatial and temporal features through a dedicated decoder and generates the final change map. Experiments on four popular datasets show that our proposal scores higher than most state-of-the-art approaches. In addition, the appealing performance is achieved with mild quantities of parameters and computations. The code is available at https://github.com/ZhengJianwei2/STENet.**

***Index Terms*— Change detection (CD), convolutional neural network (CNN), remote sensing (RS), transformer.**

I. Introduction

CHANGE detection (CD) is a crucial task in monitoring environmental variations on the surface of the Earth. The main objective lies in quantitatively analyzing the evolution within the fixed geographical location of two remote sensing (RS) images captured at different times. In recent years, with the continuous advancement of satellite technology and optical imaging sensors, RS CD has been widely applied across various domains, including disaster response [1], urban development monitoring [2], agricultural land changes [3], and water resource control [4].

Traditional CD techniques primarily involve arithmetic computations, image transformations, and classification-based solutions. Arithmetic computations derive the result map by pixelwise comparison of two images, followed by a filtering step given a preset threshold. Image transformation methods map the images to a feature space and analyze them to identify changing pixels, such as principal component analysis (PCA), multivariate change detection (MAD), and change vector analysis (CVA). With the evolution of machine learning, classification-based solutions such as decision trees [5], random forest [6], and support vector machines [7] have gained prominence. These treatments transform CD into a classification problem by extracting grown-up features, thus improving detection accuracy. Nevertheless, while traditional methods relying on manually extracted features excel in simple scenarios, their effectiveness diminishes in complex situations because of a heavy reliance on empirical design.

More recently, CD practices have witnessed significant improvement, courtesy of the potent representation capabilities offered by deep neural networks (DNNs). Deep learning-based approaches outshine traditional methods by learning features that discern changes from a vast pool of supervised samples. For instance, Chen et al. [8] reformulate convolutional neural network (CNN) features as semantic tokens, making an attempt to learn contextual information through the use of a transformer encoder. Subsequently, the learned features are embedded in pixel space via a transformer decoder (TD), resulting in satisfactory efficiency and efficacy. However, during the encoding stage, bitemporal RS features are simply aggregated by stacking operation along the channel direction, allowing limited interaction between paired images. Li et al. [9] propose a multiscale convolution channel attention to enhance the distinguishability of spatial information and minimize redundancy, followed by utilizing a bidirectional long short-term memory to acquire the temporal difference features of bitemporal images. Though this approach delves deep into the feature information of individual CD images, it lacks semantic communication between two inputs. Jiang et al. [10] leverage multikernel channel alignment attention to capture the remote semantic information of

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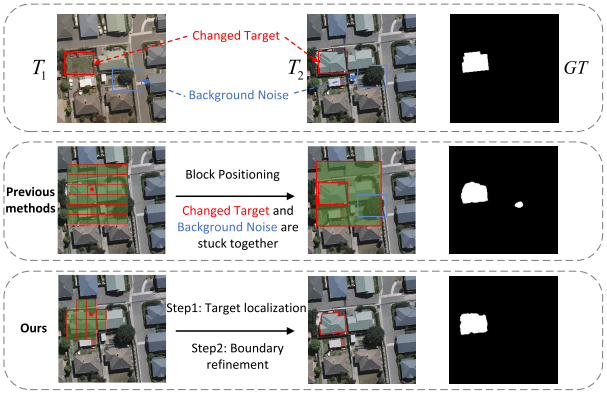


Fig. 1. Overview of the motivation for our method. Traditional methods often search for areas of change within the confines of rectangular blocks, which makes them susceptible to interference from background noise unrelated to the task. Our approach, however, initially locates change targets by dynamically selecting where to focus attention and then refines the boundaries through a TE branch. This effectively reduces the interference from background noise and alleviates the issue of blurred boundaries.

features at different scales and utilize a feature difference reconstruction module to effectively filter change information, thereby improving the perceptibility of the change region. Yet, the deficiency of context aggregation in CD images can be alleviated to some extent; this approach could not guide the attention to focus more on the change regions.

Despite the notable advancements realized through deep learning-based methodologies, several challenges endure. First, confronted with extensive datasets encompassing complex and extraneous interference, i.e., seasonal turnover, environmental differences, and illumination variations, current CD methods falter in the extraction of semantic correlations between bitemporal images. In contrast, delving too deeply into the analysis of individual images can detrimentally impact detection outcomes. Consequently, a prerequisite arises for a methodology facilitating the interaction between the semantic features of bitemporal images, effectively aggregating contextual information from bitemporal perspectives. Second, with a focus on steering attention toward sparsely changed elements while mitigating the impact of extraneous interference, it is advocated for the incorporation of sparsity into the network. This integration ensures that the CD network selectively emphasizes regions undergoing genuine changes. Nevertheless, traditional attention learning encounters substantial computation and memory challenges, particularly given long sequences. In contrast, sparse attention proves to be instrumental in significantly alleviating computational demands. Third, prevailing methods often prioritize uniquely global localization of changes or concentrate on capturing local features. This emphasis sometimes renders the prediction of changed regions vague and inaccurate. Hence, there exists a necessity to investigate alternative methods that delve into the change evolution, aiming for a more accurate identification of the intrinsic features. To alleviate the above problems, we turn to design a spatial selection and temporal evolution network (STENet) with a simple yet effective architecture. The core motivation of our proposal is illustrated in Fig. 1. Our main contributions are summarized as follows.

1) To take advantage of the sparsity inherent in changes, we elaborate dynamic selective attention (DSA) to optimize both computational efficiency and memory usage. Concretely, this mechanism steers attention toward the most semantically relevant regions by transitioning from identifying such regions at a coarser granularity to a finer one.
2) Considering global localization in spatial variation and local representation in temporal evolution, we engineer a dual-pathway architecture based on fed pairs and augmented sequences. The former utilizes DSA for spatially interactive and accurate localization, while the latter learns the detailed features via 3-D convolution blocks from pseudovideos generated from data augmentation.
3) Quantitative and qualitative experiments on four datasets demonstrate that our proposal consistently outperforms other state-of-the-art methods, especially in cases with limited samples available. Codes are available as an attachment and will be publicly released on GitHub.

II. Related Work

A. Sparse Attention Mechanism

Unlike CNNs with limited receptive fields, vision transformers [11], [12] can capture long-range dependencies, holding a global receptive field. However, such an advantage comes at the price of heavy computational complexity, which grows quadratically with the length of the input sequence. Furthermore, transformers attend to each position of the input sequence globally, which may result in redundancy and unnecessary computations. To tackle these challenges, various sparse attention mechanisms have been incorporated into vision transformers, permitting attention to be focused only on small subsets of key–value pairs rather than the entirety at once. In the realm of sparse attention, Swin transformer [13] marks a milestone with a window-based sliding operation, effectively confining self-attention (SelfAtt) calculations within nonoverlapping local regions. Similarly, Dong et al. [14] developed a cross-window mechanism, which performs attention computations in horizontal and vertical stripes within windows in parallel, thus reducing computation. However, these attention-based methods manually define the shape of the attention window, which does not allow for flexible and dynamic querying of irregular window targets.

In CD, an imbalance in the ratio of foreground to background exists, with the proportion of foreground being significantly less than background noise and the truly changed building regions being even more sparing. Thus, imposing a sparsity prior on attention is a natural choice, which could exclude irrelevant background disturbances and concentrate attention on truly changing areas. In previous studies, Sun et al. [15] considered the change sparsity as a handcrafted prior and proposed an unsupervised framework based on structural consistency, which maps one image to another and explores the correlation between different data distributions. Zhang et al. [16] leveraged the denoising capability of sparse feature

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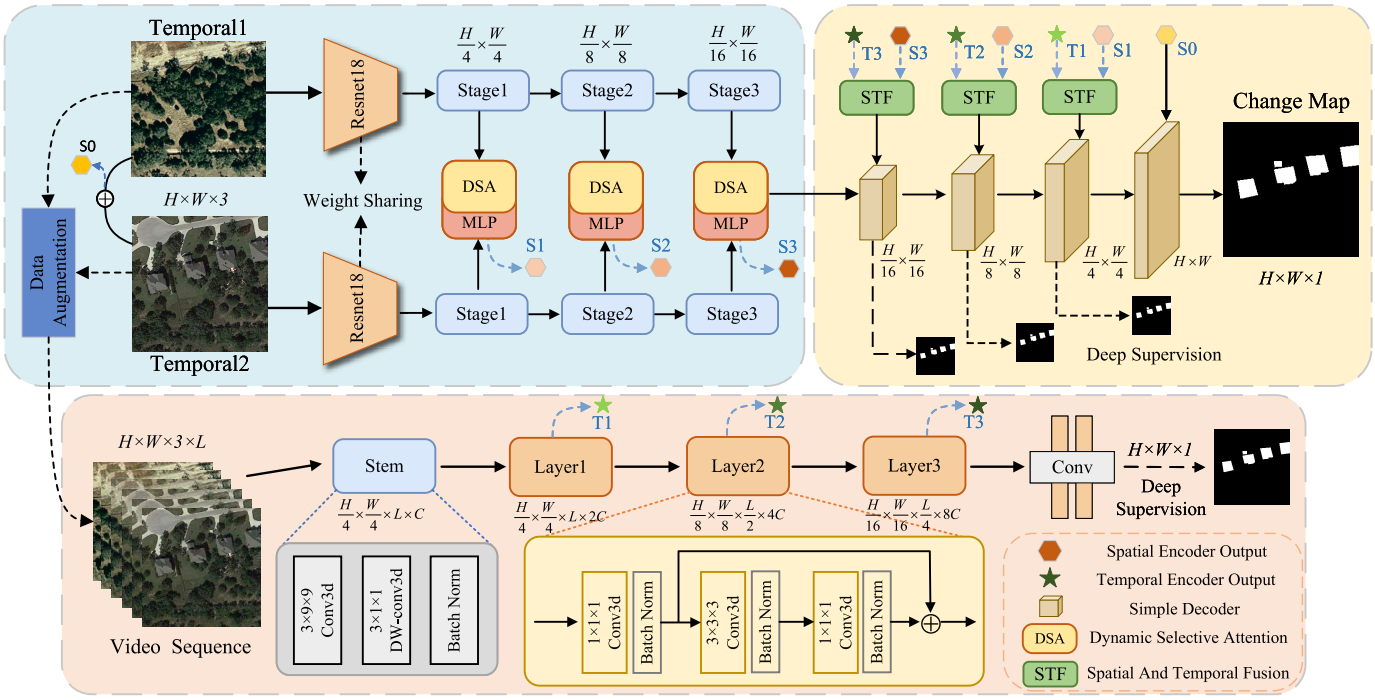


Fig. 2. Overall architecture of STENet, which mainly consists of two parallel branches, i.e., an SE and a TE, for the acquisition of joint temporal–spatial representations. The former leverages DSA for information coupling between bitemporal images, in which the focused attention is steered toward more changes. The latter explores data-augmented video sequences to enrich temporal features and sense more refined boundaries of change regions. Then, multiscale representations of the two encoders are first aggregated in feature dimensions by a simple fusion module, and finally, the fusions are decoded by progressively upsampling the features.

learning, adaptively learning representations of sparsity to distinguish between changed and unchanged pixels' semantic features. Li et al. [17] put forward a difference-enhanced pyramid pooling and a dynamic sparse attention mechanism that emphasizes pertinent changes and filters out irrelevant ones, thereby capturing precise boundary information.

While these works employ the sparsity of changing features from various aspects, their practical application still faces certain challenges, mainly the lack of constraints on unwanted background and noise, culminating in insufficient robustness against nonintrinsic changes. We attempt to investigate query awareness for attention computation, dynamically searching for semantic associations between dual-temporal images, thus achieving query-adaptive sparsity and eliminating interference from irrelevant backgrounds.

B. Deep Learning-Based CD Methods

Benefiting from the rapid development of computer vision and the availability of large-scale labeled datasets, numerous supervised CD methods on RS images have been designed and become the dominant solutions in recent years [18], [19], [20]. The majority of them regard CD as a binary segmentation task, which detects changes by classifying the extracted bitemporal features into distinct classes. Yet, this design limits semantic interactions between bitemporal images. Recently, efforts have been shifted to designing advanced network architectures and achieving semantic aggregation between bitemporal images. Representatively, the concept of fully convolutional networks was initially proposed in [21], with the aim of detection through concatenated encoders and early fusion. This architectural innovation reduces the network parameters, yet it is sensitive to background noise due to the pure employment of convolutions. Subsequently, Peng et al. [22] developed a new architecture based on U-Net++, adopting multilayer dense skip connections to generate multiscale feature maps and also using multiside output fusion (MSOF) to further enhance spatial details. However, this method prematurely merges the semantic information of bitemporal images, leading to a risk of detail loss during long-distance modeling.

More recently, widespread interest in transformers has also sparked a revolution in the CD field. Chen et al. [8] were pioneers in combining transformers with CNN within the CD framework, using ResNet to map bitemporal RS images to different feature maps, and then enriching the spatiotemporal context with vanilla transformer modules. The generally satisfactory outcomes validated the success of combining CNNs with transformers. Moreover, to focus more on the learning of change areas, Jiang et al. [23] introduced an efficient multidimensional attention network (MANet) that maintains an excellent computational capacity for differential attention while also achieving remarkable feature aggregation. Xu et al. [24] presented a transformer-based contextual aggregation network, utilizing a progressively sampled visual transformer and effectively eliminating interference from irrelevant changes. These methods have successfully aggregated the semantic features between bitemporal images, achieving commendable results, yet global attention computation still entails significant computational costs. In RS images, change areas are typically smaller than unchanged ones, and the changes are naturally sparse. Therefore, how to focus limited computational resources on the areas of real change remains a challenge.

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III. Methodology

In this section, the details of our STENet are discussed. The overall flowchart is shown in Fig. 2. Generally, the model follows a dual-branch architecture and mainly consists of a spatial encoder (SE), temporal encoder (TE), and a simple fusion decoder. In the sequel, we first discuss the overview framework of STENet, followed by a sequential presentation of three modules. Finally, the loss function is clarified.

A. Network Overview

To provide a desirable solution for the CD task on RS images, Fig. 2 hammers at exploring the feature interactions from both spatial and temporal perspectives. Within the spatial domain, the original bitemporal images T_1 and T_2 serve as inputs. An SE is meticulously constructed to extract contexts between the bitemporal images, aiding in the localization of changing regions. For simplicity, the inputs are individually extracted by two parallel branches of a siamese pretrained CNN (ResNet18), generating feature maps with spatial sizes of 64×64, 32×32, and 16×16, respectively. Furthermore, the extracted multilevel maps undergo DSA for feature coupling between the image pairs, which collectively steers attention to regions with significant variations. On the temporal side, we employ a mix-up augmentation scheme to produce a temporal evolution video from the given two inputs and then enrich the temporal features through a series of improved ResNet-3-D layers. Finally, a spatial and temporal fusion (STF) module is designed to integrate the outputs of both spatial and temporal branches, whose results are fed into a simple decoder for layer-by-layer feature fusion and upsampling, achieving the final predictions.

B. Dynamic Selective Attention

Given the nature of capturing CD across multiple spatiotemporal shots, irrelevant interference often spreads all over the acquired images. Furthermore, when dealing with imbalanced foreground-background targets, identifying changes becomes notably challenging. To address these issues, we propose a dynamic selection mechanism to distribute attention over regions that truly undergo temporal changes.

Specifically, in Fig. 3, given 2-D dual-time feature maps X_1, $X_2 \in \mathbb{R}^{H \times W \times C}$ achieved from backbone network, we first perform a linear projection and reshape them to obtain the queries (Q_n), keys (K_n), and values (V_n) with $n = \{1, 2\}$, which are all of size $S^2 \times (H/S) \times (W/S) \times C$. S is a hyperparameter representing the number of regions divided by our method. Then, we chunk them into S^2 feature regions, each with the shape of $(H/S) \times (W/S) \times C$. Afterward, these smaller regions are taken as the units for interaction between bitemporal images. The concrete structure of DSA further contains two subassemblies including rough selective attention (RSA) and fine-grained pixel attention (FPA).

1) Rough Selective Attention: To simplify the calculation of interaction attention and determine the focal regions for each area, a two-step process is implemented. Initially, spatial dimension reduction is performed on both Q_1 from X_1 and K_2 from X_2, averaging the height and width dimensions of the tensors to obtain the mean region features. This eliminates the length and width dimensions of the original features Q_1 and K_2, resulting in the dimension-reduced $Q_1^m \in \mathbb{R}^{S^2 \times C}$ and $K_2^m \in \mathbb{R}^{S^2 \times C}$

$$Q_1^m = \text{Mean}_{\dim(H,W)}(Q_1) \tag{1}$$

$$K_2^m = \text{Mean}_{\dim(H,W)}(K_2). \tag{2}$$

Subsequently, an adjacency matrix $A_r \in \mathbb{R}^{S^2 \times S^2}$ that denotes the rough region-similarity scores between Q_1 and K_2 can be obtained, which is simply computed by matrix multiplication of Q_1^m with K_2^m, due to its efficacy and prevalence

$$A_r = Q_1^m \times \left(K_2^m\right)^{\mathrm{T}} \tag{3}$$

where T represents a transpose operation. The entries in the adjacency matrix A_r measure the semantic relationship between two regions, represented by the attention score level. Based on the adjacency matrix, we can easily identify regions with high similarity, essentially pinpointing the areas where actual change is taking place. The next core step then lies in filtering the matrix by selecting only the top-k regions among the S^2 units on Key K_2^m, which are most semantically related to the Query Q_1^m. The formula is given as follows:

$$I_r = \text{TopkIndex}(A_r). \tag{4}$$

$I_r \in \mathbb{R}^{k \times S^2}$ represents the indices of the top-k strongest regions that are dynamically selected. In other words, the regions that require the most attention on the original feature map can be easily located based on the position information provided by the indices.

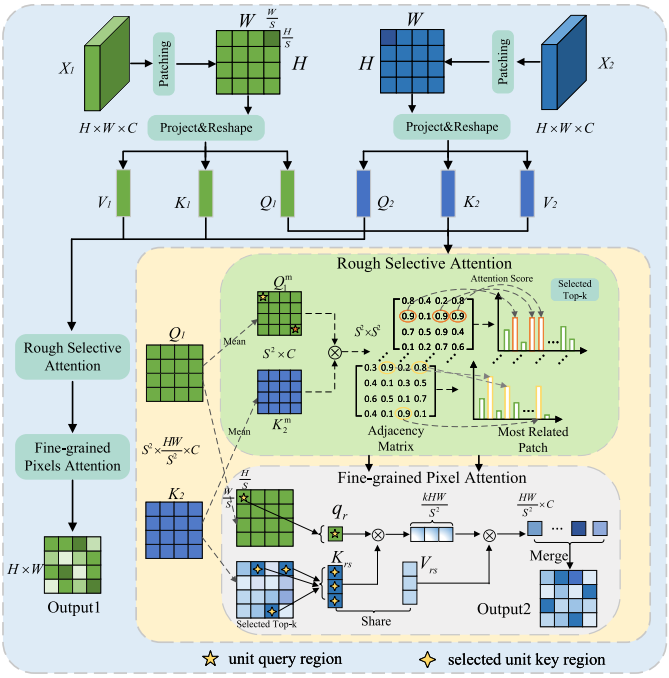


Fig. 3. Overview of DSA. First, in RSA, with spatial dimensionality reduced, an affinity matrix A_r is computed using query Q_1 and key K_2, followed by the obtainment of top-k selected regions based on attention scores in A_r. Then, in FPA, for each unit query q_r, the token-to-token attention is achieved with the set of the selected key–value pairs. Finally, the overall attention is obtained by stitching the outputs of all S^2 regions together.

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2) Fine-Grained Pixel Attention: Through the index I_r between queries and keys, the precise attention values can be learned using fine-grained interpixel connections. In this instance, we start from a single query region q_r on Q_1, the smallest unit of interaction between bitemporal images, to explore how each small region on Q_1 calculates attention with the corresponding top-k key–value pairs on K_2.

Specifically, we divide $Q_1 \in \mathbb{R}^{S^2\times(H/S)\times(W/S)\times C}$ into S^2 unit regions, namely, $Q_1 \rightarrow \{q_1, q_2, \ldots, q_{S^2}\}$. On that basis, the single query $q_r \in \mathbb{R}^{(HW/S^2)\times C}$ for each unit region can be achieved

$$q_r = \frac{Q_1}{S^2}, \quad r \in (1, 2, \ldots, S^2). \tag{5}$$

Note again that for each query q_r, the top-k key–value pairs selected in K_2 can be located given the concerned index I_r. Subsequently, we aggregate these key–value pairs into a long tensor with a shape of $(kHW/S^2) \times C$, resulting in K_{rs} and V_{rs}

$$K_{\text{rs}} = \text{Aggregate}(K_2, I_r) \tag{6}$$

$$V_{\text{rs}} = \text{Aggregate}(V_2, I_r). \tag{7}$$

Following this, the cross-attention (CrossAtt) between q_r, K_{rs}, and V_{rs} is computed as follows:

$$\begin{aligned}\text{Att}_r &= \text{CrossAtt}(q_r, K_{\text{rs}}, V_{\text{rs}}) \\ &= \text{Softmax}(q_r \cdot K_{\text{rs}}^{\text{T}}) \cdot V_{\text{rs}} \\ &= \left(\frac{HW}{S^2} \times C\right) \times \left(\frac{kHW}{S^2} \times C\right)^{\text{T}} \times \left(\frac{kHW}{S^2} \times C\right)\end{aligned} \tag{8}$$

where $\text{Att}_r \in \mathbb{R}^{(HW/S^2)\times C}$. Softmax is a commonly used activation function. With the attention for each unit obtained, we then merge the attention of all S^2 regions and get the output $O_1 \in \mathbb{R}^{H\times W\times C}$

$$O_1 = \text{Merge}\{\text{Att}_1, \text{Att}_2, \ldots, \text{Att}_{S^2}\}. \tag{9}$$

For O_1, we establish the interactive attention distribution by using query Q_1 from the feature map X_1 and key K_2 from the feature map X_2. The concerned attention is focused more on X_2, potentially leading to some information loss. Therefore, the same parallel computation is applied, involving Q_2, K_1, and V_1, deriving O_2 that supplements necessary information. Hence, the attentions O_1 and O_2 are jointly learned.

3) Complexity Analysis of DSA: The computation of DSA mainly consists of three parts: linear projection, RSA, and FPA. The total amount of computations is, therefore,

$$\begin{aligned}\text{FLOPs} &= \text{FLOPs}_{\text{proj}} + \text{FLOPs}_{\text{rough}} + \text{FLOPs}_{\text{fine}} \\ &= 6HWC^2 + 4\left(S^2\right)^2 C + 4\frac{k(HW)^2}{S^2}C \\ &= 6HWC^2 + 2C\left(2S^4 + \frac{k(HW)^2}{S^2} + \frac{k(HW)^2}{S^2}\right) \\ &\geq 6HWC^2 + 6C\left(2S^4 \cdot \frac{k(HW)^2}{S^2} \cdot \frac{k(HW)^2}{S^2}\right)^{\frac{1}{3}} \\ &= 6HWC^2 + 6Ck^{\frac{2}{3}}(2HW)^{\frac{4}{3}}\end{aligned} \tag{10}$$

where C is the channel number of the feature map and k is the number of regions involved in the computation. The inequality in (10) holds when $2S^4 = (k(HW)^2/S^2)$, which means that when we select a moderate number of divided regions S^2, DSA enjoys a computational complexity of $O(HW)^{(4/3)}$, which is surely less than ordinary attention complexity $O(HW)^2$.

C. Three-Dimensional Temporal Evolution Module

Diverging from SEs that extract features across different regions, the TE seeks to augment the model with richer information along the channel dimension. Intuitively, the change between the front and back images is similar to the temporal variation in different states among frames in a video. Therefore, we turn to transform the CD problem into a task of video understanding, in which the crucial step lies in probing into an evolution process, or, more precisely, a sequence of frames generated from a given pair of bitemporal images. Since collecting more data for a genuine and high-quality transition video is challenging and requires substantial resource costs, we propose an alternative strategy that directly produces frame sequences from dual-time images, without the need to manually acquire external data. Concretely, a mix-up data augmentation scheme is employed to gradually transition from the initial frame (T_1) to the counterpart (T_2) by interpolating to generate intermediate frames, resulting in a continuous pseudovideo frame tensor that consists of N frame. That is to say, using the original images, T_1 and T_2, along with the frame index n, we deduce the nth $(0 \leq n < N)$ frame F_n as

$$F_n = (1-\mu)T_1 + \mu T_2, \quad \mu \in [0, 1] \tag{11}$$

where $\mu = n \times \text{step}$ with step set as $(1/N - 1)$ represents the temporal variation coefficient. An example of the constructed pseudovideo sequence is provided in Fig. 4.

In (11), we weight two frames to achieve a sequence of blended images. As μ increases, the blending frame gradually transitions from being close to the first input to the second one. It is clear that when $n = 0$ and $\mu = 0$, then frame $F_{n=0}$ is equivalent to the original image T_1. While at $n = N - 1$, $\mu = 1$, the frame $F_{n=N-1}$ corresponds to the original image T_2. So far, we have successfully constructed a simple but efficient pseudotemporal video from the input images. Note again that within the video frames, the truly changing regions will present a gradual evolution process, yet the unchanged ones bear trivial impacts from the mix-up data augmentation.

As shown in Fig. 2, the TE consists of a stem module and three progressive 3-D ResNet layers. In order to steer the attention more to temporal variations, it is required to minimize the interference from spatial disturbance. To that end, the stem module endeavors to optimize the extraction of temporal features and, meanwhile, reduce the redundant spatial information. Specifically, we introduce a 3-D convolution at the onset of the TE. The convolution kernel is configured at $3 \times 9 \times 9$, and the stride is determined as $1 \times 4 \times 4$. On that basis, a depth-separable convolutional layer follows, featuring a $3 \times 1 \times 1$ kernel and a stride of 1. This layer executes spatial convolution by individually filtering each

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(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 4. Example of synthesizing pseudo-video frames, where (a) and (h) represent the original images T_1 and T_2 respectively, and (b)–(g) represent the intermediate transition frames.

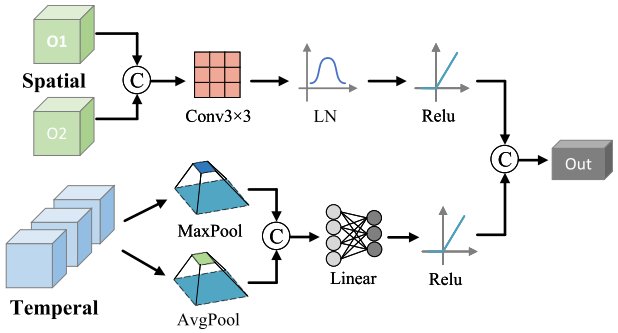


Fig. 5. Overview of the STF module.

input channel, followed by a point-by-point convolution of the results. The overall process effectively reduces the parameter count while maintaining an individual treatment for each channel. Consequently, a four times downsampled feature sequence is obtained, facilitating subsequent layers to process higher level semantic features.

To progressively capture the temporal feature map information from different channel configurations in pseudovideo sequences, three 3-D bottleneck residual blocks are consecutively employed. For each block, an initial $1\times1\times1$ convolution layer serves to reduce the number of input channels, followed by an intermediate $3 \times 3 \times 3$ convolution to extract temporal information. The final stage involves a secondary $1 \times 1 \times 1$ convolution, aiming at restoring the channel count. Throughout this process, we maintain a constant channel count for the pseudovideo length, a strategic design that not only lowers computational costs but also allows for the economical construction of broader and deeper networks. Furthermore, deep supervision is imposed on the output of the third residual block to enhance the discrimination of temporal features. Ultimately, the resulting scale-varying temporal feature sequence is fed into a fusion module, where it interacts with spatial features obtained from another branch.

D. STF Module

The devised dual-encoder parallel architecture serves the purpose of enhancing semantic correlations between bitemporal images in both temporal and spatial aspects. For the postextraction of rich feature information from diverse perspectives, the consideration shifts to how information from the two encoders can synergistically interact for mutual enhancement. As indicated by [25] and [26], interpathway information interaction proves instrumental in advancing network prediction accuracy. Consequently, illustrated in Fig. 5, a spatial-temporal fusion (STF) module is introduced. Specifically, the initial step involves channelwise fusion of two outputs from the SE, denoted as O_1 and O_2, both within $\mathbb{R}^{H\times W\times C}$. In addition, a 3×3 convolutional layer orchestrates the restoration of channel dimensions to the original size C. Following this, layer normalization (LN) is further applied, ensuring the model stability during training, followed by rectified linear units (ReLU) that ensure necessary nonlinear features. This process can be formally expressed as

$$O_{\text{Spatial}} = \text{Relu}(\text{LN}(\text{Conv}(\text{Cat}(O_1, O_2)))). \tag{12}$$

Concurrently, the TE produces 3-D features denoted as $O_t \in \mathbb{R}^{L\times H\times W\times C}$. To transform O_t into 2-D features for subsequent spatial–temporal feature interaction, the operations of global average pooling and global max pooling are first performed along the temporal dimension L of the 3-D features. This treatment seeks to distill critical information for each temporal frame, shedding light on the specific locations where true changes occur. The resulting features are then concatenated along the channel dimension, generating feature maps with channel dimension $2C$. Afterward, the merged feature map is further processed through a fully connected layer to restore the channels to the original size C. The equations are described as follows:

$$O_{\text{Temporal}} = \text{Linear}(\text{Cat}(\text{MaxPool}(T), \text{AvgPool}(T))). \tag{13}$$

Finally, the spatial feature map and the temporal feature map culminate in an aggregation along the channel direction through the application of a 3×3 convolutional layer, complemented by LN and ReLU activation, yielding the ultimately fused feature. The description is given as follows:

$$O_{\text{fusion}} = \text{Conv}(\text{Cat}(O_{\text{Spatial}}, O_{\text{Temporal}})). \tag{14}$$

E. Hybrid Loss Function

We utilize the deep supervision mechanism of change prediction to train the overall model.

1) Deep Supervision in Decoding: In most cases, the proportion of changed regions is significantly smaller than the unchanged ones, resulting in a class imbalance problem. To guide a better prediction of both results and intermediate outputs, we employ a hybrid loss consisting of a binary cross-entropy (L_{bce}) and a dice (L_{dice})

$$L_{\text{pre}} = \sum_{i=0}^{S} L_{\text{bce}}(\text{Pre}_i, G) + L_{\text{dice}}(\text{Pre}_i, G) \tag{15}$$

where Pre_i is the prediction for ith stage in the total of S stages and G represents the ground truth.

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2) Deep Supervision in Encoding: We investigate the influence of constraining L_{tem} on the 3-D-ResNet module during temporal feature extraction. This approach aims to expedite the convergence of the TE, leading to a well-defined contour of the task-relevant target. To ensure the stability of the gradient during backpropagation, we adopt the mean absolute error (MAE) as the chosen loss function

$$L_{\text{tem}} = L_{\text{mae}}(\text{Pre}_{\text{tem}}, G). \tag{16}$$

The overall supervision loss is then $L_{\text{total}} = L_{\text{pre}} + \lambda L_{\text{tem}}$.

IV. Experiments

A. Experimental Protocol

Extensive experiments are conducted on four representative CD datasets to validate the practical performance of the proposed STENet.

1) The cropland change detection dataset (CLCD-CD) [3] is constructed for the analysis of cropland change, focusing on detecting changes related to agricultural land, buildings, roads, and so on, and consists of 600 RGB image pairs with a size of 512 × 512. The bitemporal images in CLCD were collected by Gaofen-2 in Guangdong Province, China, in 2017 and 2019, respectively, with spatial resolution ranging from 0.5 to 2 m. Due to the limited GPU memory, we cut the original image pairs into 256 × 256 patches. In total, CLCD applies 1440/480/480 pairs of patches for training/verification/testing sets, respectively.
2) The Guangzhou dataset (GZ-CD) [27], mainly involving building targets, is a VHR satellite image dataset with a spatial resolution of 0.55 m/pixel, which covers the suburb areas of Guangzhou City, China, from 2006 to 2019. Through the Google Earth service, a total of 19 season-varying image pairs with spatial sizes ranging from 1006 × 1168 to 4936 × 5224 are obtained. Consistent with the processing of most previous treatments, we uniformly cut the inputs to size 256 × 256. These image patches are further divided into 2504/313/313 pairs for training/verification/test sets, respectively.
3) The Wuhan university building change detection dataset (WHU-CD) [2] records the construction of buildings (postearthquake) in the region of Christchurch, New Zealand, after a magnitude 6.3 earthquake in February 2011. This dataset consists of two aerial images of size 32 507 × 15 354 and a resolution of 0.2 m/pixel taken over the same area in 2012 and 2016. The WHU dataset focuses exclusively on building changes. As no official split was suggested in [2], we first cropped the original images into 256 × 256 chips without overlap and then randomly split the chips into three subsets: 6096/762/762 for training/validation/test, respectively.
4) The Sun Yat-Sen University dataset (SYSU-CD) [28] includes 20 000 orthographic aerial pairs with a spatial size of 256 × 256 and a spatial resolution of 0.5 m, which are originally gathered in Hong Kong. It is noteworthy that SYSU-CD covers various types of targets in addition to buildings, such as vessels, roads, lakes, and vegetation, presenting a tougher ordeal for CD tasks. For training/validation/testing sets, we allocate 12 000/4000/4000 samples, respectively.

Five metrics are used to assess the direct detection performance, i.e., precision (Pre), recall (Rec), $F1$-score ($F1$), intersection over union (IOU), and overall accuracy (OA). $F1$ is the harmonic mean of Pre and Rec, where Pre represents the false positive rate and Rec reflects the leakage rate. These metrics are defined as

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP}) \tag{17}$$

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}) \tag{18}$$

$$\text{F1} = 2/(\text{Recall}^{-1} + \text{Precision}^{-1}) \tag{19}$$

$$\text{IOU} = \text{TP}/(\text{TP} + \text{FP} + \text{FN}) \tag{20}$$

$$\text{OA} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN}) \tag{21}$$

where TP stands for the number of true positives, TN represents the number of true negatives, FP is the number of false positives, and FN denotes the number of false negatives.

The implementation of STENet is carried out on the PyTorch framework, which has undergone rigorous experimentation for robustness and fairness considerations. All experiments are systematically conducted on a dedicated platform featuring two NVIDIA RTX 3090 GPUs, each boasting 24 GB of video memory. Practically, we set the number of constructed video frames N to 8 and the weighting coefficient of the auxiliary loss λ to 0.5. Adhering to established practices, we employ various data augmentation techniques during training, including flipping, rescaling, cropping, and Gaussian blurring. These augmentations collectively enhance the model's robustness and contribute to the overall reliability of our experiments. For optimization, we leverage stochastic gradient descent (SGD) with an initial learning rate of 1e-2, a momentum of 0.99, and a weight decay of 5e-4. The chosen batch size and epoch are respectively set as 8 and 200. Note that the learning rate undergoes a linear decrease as the epochs progress. A validation step is carried out after each epoch to select the model with the highest $F1$. Then, the final performance is reported on the test subset.

B. Comparison to State-of-the-Arts

We compare the efficacy and efficiency of our approach with 18 previously reported state-of-the-art competitors, including five classic convolutional-based methods commonly used in the CD field, i.e., fully convolutional early fusion (FC-EF) [21], fully convolutional siamese-concatenation (FC-Siam-conc) [21], fully convolutional siamese-difference (FC-Siam-diff) [21], fully residual network (FresUNet) [29], and edge-guided recurrent CNN (EGRCNN) [20], and seven attention-based methods, i.e., SNUNet [18], bitemporal image transformer (BIT) [8], intrascale cross-interaction and interscale feature fusion network (ICIFNet) [30], ChangeFormer [31], densely attentive refinement network (DARNet) [19], scale and relation aware siamese network (SARASNet) [32], and w-shaped hierarchical network (WNet) [33]. Six recently reported CD methods based on multiscale feature fusion

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TABLE I

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON CLCD-CD AND GZ-CD DATASETS. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Methods	Params	FLOPs	CLCD					GZ				
			Pre(%)	Rec(%)	F1(%)	IOU(%)	OA(%)	Pre(%)	Rec(%)	F1(%)	IOU(%)	OA(%)
FC-EF$_{18}$	1.35	3.57	65.86	53.29	58.91	41.75	94.47	81.29	68.61	74.41	59.25	95.74
FC-Siam-Di$_{18}$	1.35	4.72	65.83	53.23	58.86	41.70	94.46	80.36	61.80	69.87	53.69	95.19
FC-Siam-Conc$_{18}$	1.55	5.32	64.96	60.33	62.55	45.51	94.62	80.37	68.97	74.23	59.03	95.68
FresUNet$_{19}$	**1.11**	2.02	66.99	67.08	67.04	50.42	95.09	87.97	77.44	82.48	70.65	97.10
EGRCNN$_{22}$	9.64	17.64	70.66	66.09	68.30	51.86	95.44	87.91	80.36	83.96	72.36	97.23
SNUNet$_{22}$	12.04	54.82	63.14	65.78	64.44	47.53	94.60	85.81	79.63	83.32	71.88	97.19
BIT$_{22}$	3.55	10.60	65.82	53.67	59.13	41.97	94.48	88.23	79.23	83.88	72.24	97.27
ICIFNet$_{22}$	23.82	25.36	**76.96**	65.09	70.53	54.47	95.95	90.17	82.35	85.65	74.91	97.53
ChangeFormer$_{22}$	29.82	11.64	71.59	58.99	65.25	48.42	95.24	89.75	78.43	83.78	73.36	97.42
DARNet$_{22}$	15.09	64.21	73.74	61.65	67.16	50.55	95.51	87.32	77.76	82.26	69.87	96.97
SARASNet$_{23}$	89.28	153.38	63.22	61.93	62.57	45.53	94.49	88.28	83.79	85.17	74.19	97.42
P2VNet$_{23}$	5.42	32.84	68.35	68.48	68.41	51.99	95.30	89.69	**84.91**	**87.49**	**78.07**	**97.74**
CTFINet$_{23}$	4.15	9.93	64.36	62.57	63.45	46.47	94.64	89.36	82.71	85.91	75.43	97.58
A2Net$_{23}$	3.78	3.01	63.85	53.68	58.33	41.17	94.19	88.44	79.99	84.01	72.43	97.25
LCANet$_{23}$	1.71	**1.98**	72.52	62.83	67.33	50.75	95.46	89.65	83.97	86.63	76.41	97.65
AFCF3D-Net$_{23}$	17.54	31.72	75.61	60.93	66.99	50.37	95.68	**92.32**	74.83	82.76	76.33	96.81
WNet$_{23}$	43.07	19.20	60.21	**67.60**	63.69	46.72	94.26	89.70	81.25	86.04	73.47	97.29
DMINet$_{23}$	6.24	14.42	**77.33**	65.79	**71.09**	**55.15**	**96.02**	89.31	83.90	86.52	76.24	97.64
Ours	10.95	14.73	75.69	**70.05**	**72.76**	**57.19**	**96.10**	**90.28**	**85.25**	**87.77**	**78.21**	**97.84**

TABLE II

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON WHU-CD AND SYSU-CD DATASETS. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Methods	Params	FLOPs	WHU					SYSU				
			Pre(%)	Rec(%)	F1(%)	IOU(%)	OA(%)	Pre(%)	Rec(%)	F1(%)	IOU(%)	OA(%)
FC-EF$_{18}$	1.35	3.57	78.79	68.15	73.08	57.58	97.61	72.28	72.85	72.57	56.94	87.01
FC-Siam-Di$_{18}$	1.35	4.72	80.44	54.59	65.04	48.19	97.20	84.94	51.45	64.08	47.15	86.40
FC-Siam-Conc$_{18}$	1.55	5.32	75.89	69.30	72.61	56.99	97.51	83.03	71.62	76.89	62.45	89.35
FresUNet$_{19}$	**1.11**	2.02	86.55	77.68	81.88	69.32	98.36	78.56	74.96	76.38	61.85	89.12
EGRCNN$_{22}$	9.64	17.64	90.23	81.09	84.86	75.03	98.67	84.36	78.41	81.64	69.02	91.64
SNUNet$_{22}$	12.04	54.82	85.58	80.99	83.22	71.26	98.44	83.49	76.37	79.77	66.35	90.87
BIT$_{22}$	3.55	10.60	86.64	81.39	83.98	72.39	98.52	78.73	75.68	77.17	62.83	89.44
ICIFNet$_{22}$	23.82	25.36	92.98	85.56	88.32	79.24	98.96	83.37	78.51	80.74	68.12	91.24
ChangeFormer$_{22}$	29.82	11.64	92.16	83.52	87.63	77.98	98.86	82.59	68.53	74.91	59.89	89.17
DARNet$_{22}$	15.09	64.21	87.22	85.19	86.19	75.74	98.70	**85.62**	77.68	81.46	68.72	91.66
SARASNet$_{23}$	89.28	153.38	93.49	83.61	88.68	79.67	98.96	84.56	77.27	80.62	67.54	91.36
P2VNet$_{23}$	5.42	32.84	**94.09**	**87.48**	**90.67**	**82.93**	**99.14**	81.07	77.18	79.08	65.39	90.37
CTFINet$_{23}$	4.15	9.93	89.48	79.07	83.95	72.35	98.56	**86.51**	71.51	78.30	64.33	90.65
A2Net$_{23}$	3.78	3.01	88.58	85.82	87.18	77.26	98.79	84.76	77.20	80.81	67.79	91.35
LCANet$_{23}$	1.71	**1.98**	93.23	86.61	89.80	81.48	99.06	84.61	79.07	81.75	69.13	**91.67**
AFCF3D-Net$_{23}$	17.54	31.72	93.28	86.93	90.36	82.42	99.13	81.25	80.83	81.04	68.13	91.08
WNet$_{23}$	43.07	19.20	92.72	84.30	88.31	79.07	98.94	83.20	79.59	81.35	68.57	91.39
DMINet$_{23}$	6.24	14.42	91.94	85.67	88.69	79.68	98.97	82.84	**81.32**	**82.07**	**69.60**	91.51
Ours	10.95	14.73	**94.21**	**88.97**	**91.51**	**84.35**	**99.21**	82.97	**81.62**	**82.29**	**69.92**	**91.68**

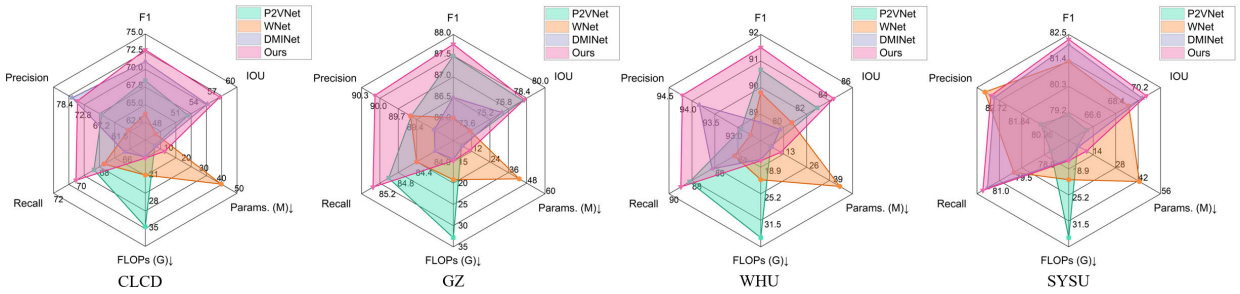


Fig. 6. Comparisons of different metrics on CLCD, GZ, WHU, and SYSU datasets.

networks are also compared, including pair-to-video network (P2VNet) [26], cross-temporal feature interaction network (CTFINet) [34], A2Net [35], lightweight collective-attention network (LCANet) [36], adjacent-level feature cross-fusion with 3-D-CNN (AFCF3D) [37], and dual-branch multilevel intertemporal network (DMINet) [25].

Note that for the competing models where pretrained models are readily available, we simply borrow them

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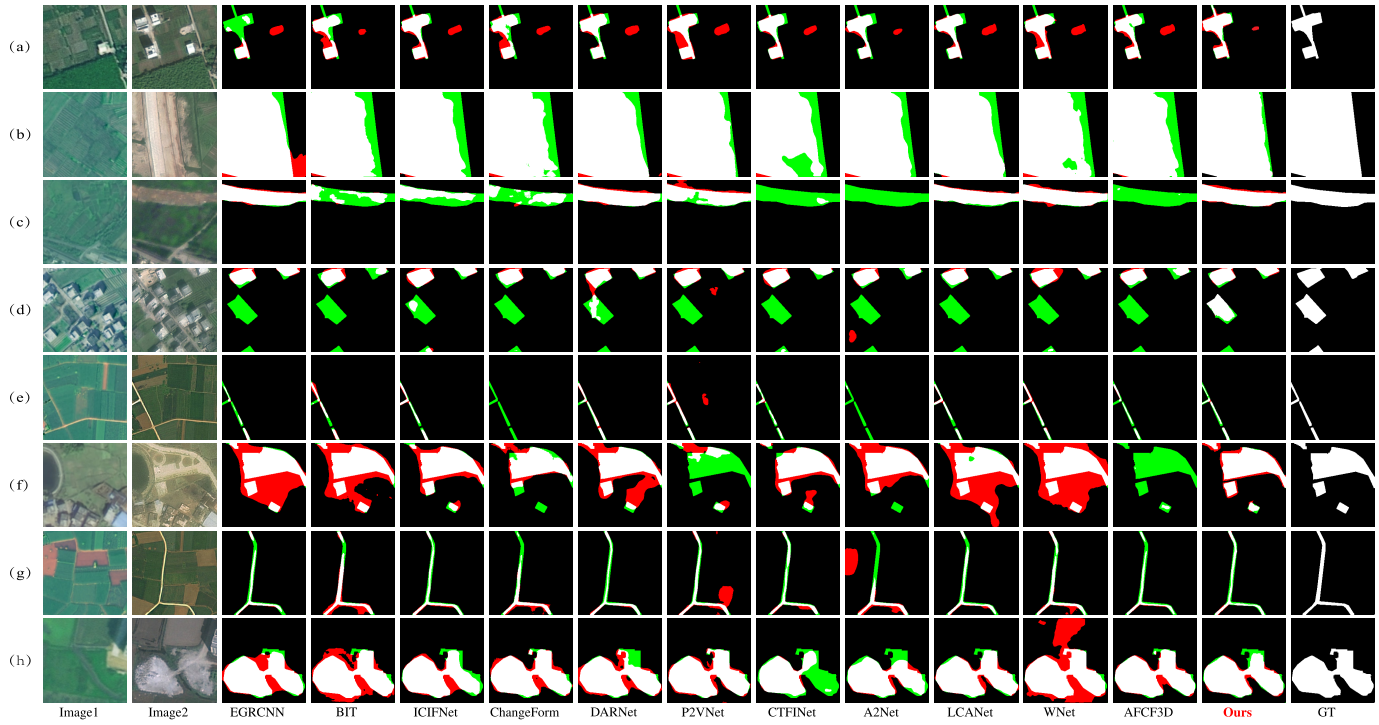


Fig. 7. Visual results of different models on CLCD-CD. (a)–(h) Eight representative samples. GT is the ground truth. In predictions, white is a true positive, black is a true negative, red indicates a false positive, and green stands for a false negative.

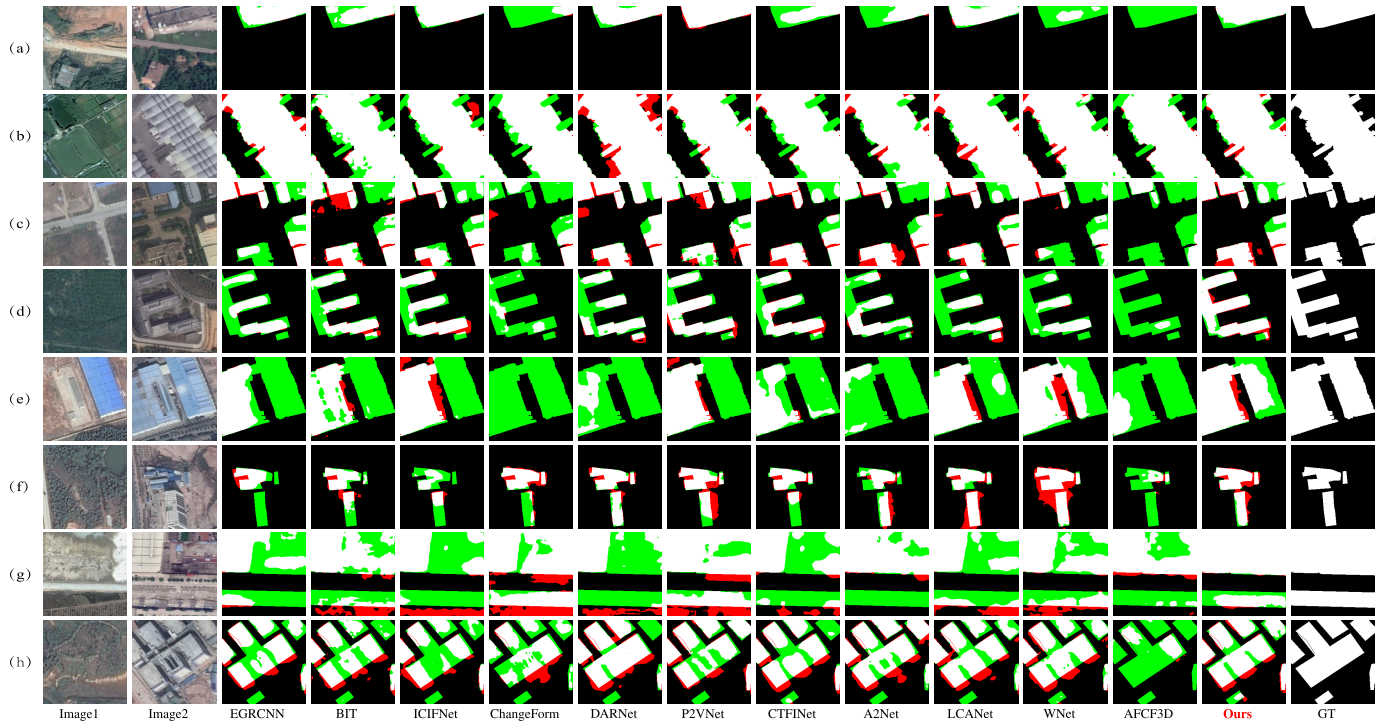


Fig. 8. Visual results of different models on GZ-CD. (a)–(h) Eight representative samples. GT is the ground truth. In predictions, white is a true positive, black is a true negative, red indicates a false positive, and green stands for a false negative.

from the original authors for validation purposes. For other competitors that no pretrained models can be easily touched, we have reimplemented these methods and achieved respectively the optimal results under the same experimental environment, with their individual hyperparameters fine-tuned.

To assess the practices of the proposed STENet model, we first conduct extensive experiments on all used RS datasets and summarize the results quantitatively. Tables I and II present the numerical results for all competing methods. In addition to five efficacy metrics (i.e., Pre/Rec/$F1$/IOU/OA), two efficiency indicators, including model parameters (Params) and computational cost (FLOPs), are also provided. It is evident that our proposed approach outperforms competitors across all benchmark datasets in terms of efficacy metrics. Specifically, against the cutting-edge

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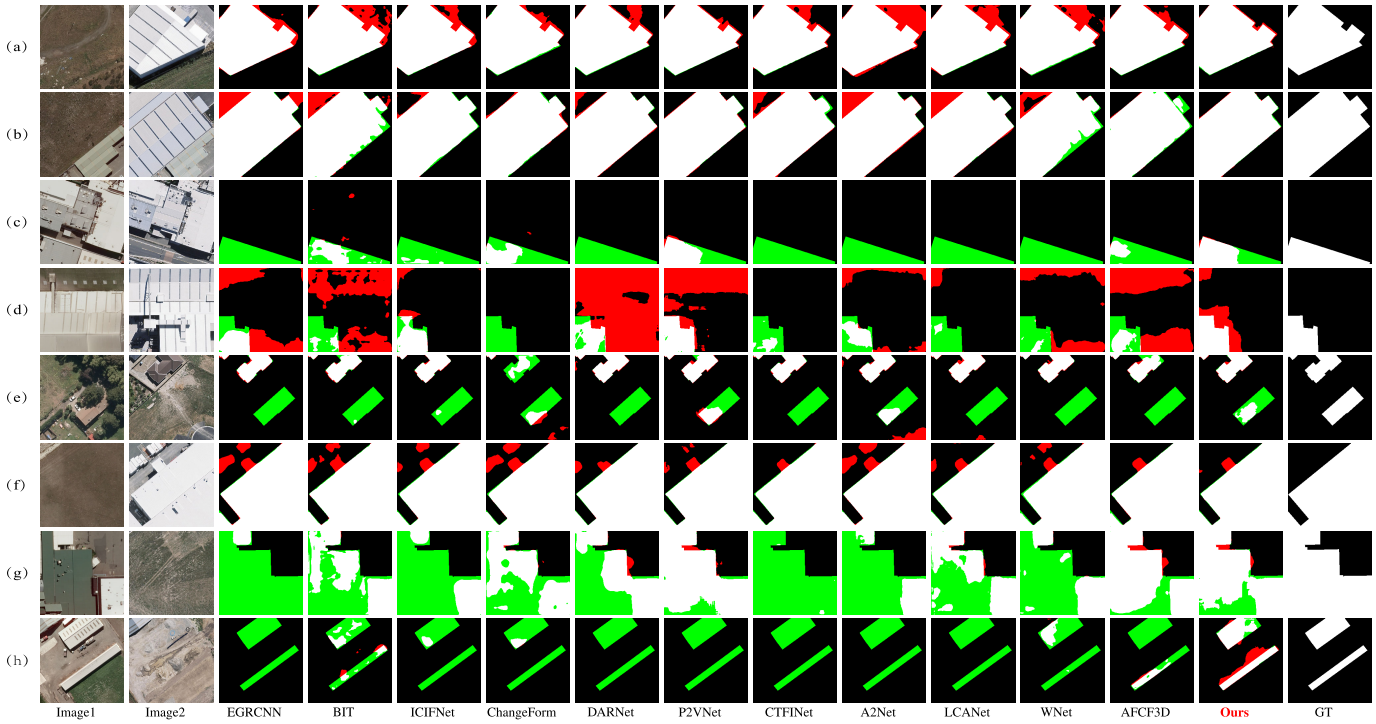


Fig. 9. Visual results of different models on WHU-CD. (a)–(h) Eight representative samples. GT is the ground truth. In predictions, white is a true positive, black is a true negative, red indicates a false positive, and green stands for a false negative.

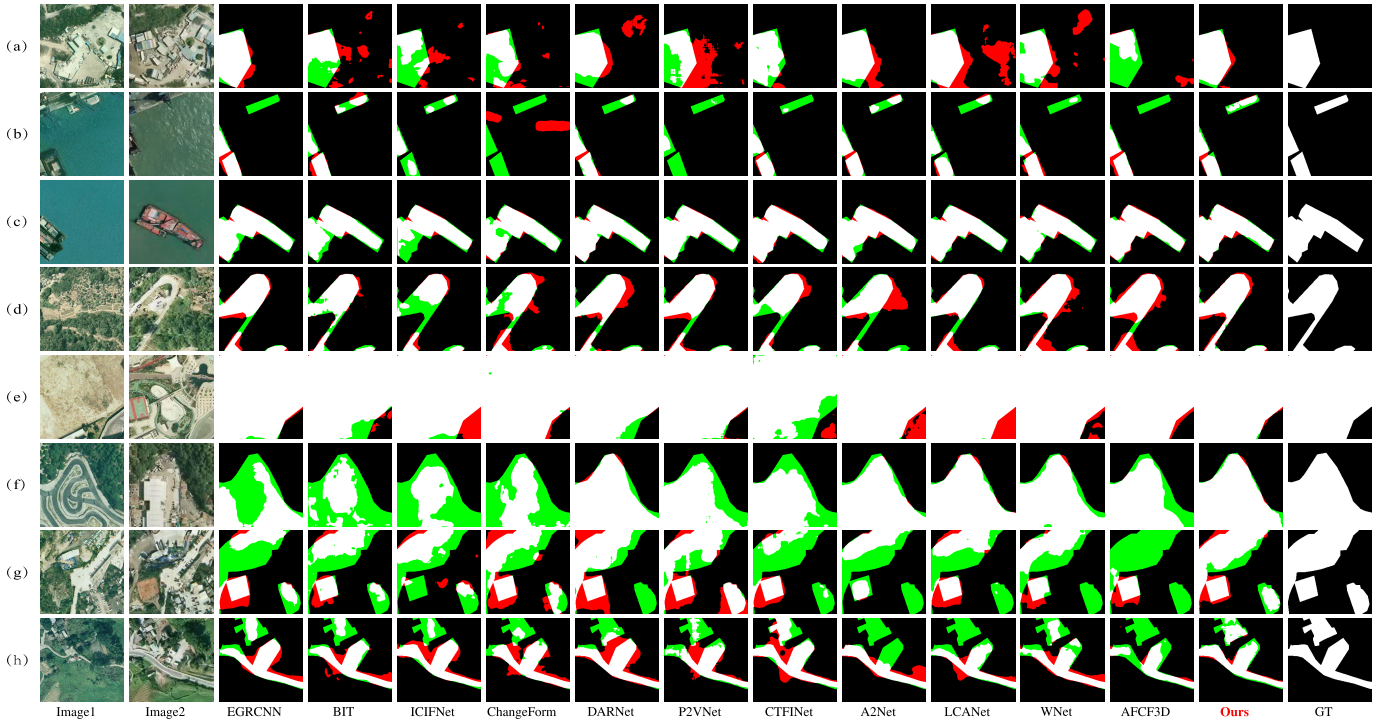


Fig. 10. Visual results of different models on SYSU-CD. (a)–(h) Eight representative samples. GT is the ground truth. In predictions, white is a true positive, black is a true negative, red indicates a false positive, and green stands for a false negative.

P2VNet, STENet achieves significant gains on CLCD and SYSU datasets, with $F1$ superiority of 4.35% and 3.21%, respectively. In comparison with DMINet, our approach demonstrates $F1$ improvements of 2.82% and 1.25% on WHU and GZ datasets, respectively. All these quantitative results indicate that modeling both spatial and temporal information can effectively enhance the localization capability for change regions. Fig. 6 further visually shows that the highest efficacy metrics are achieved with mild FLOPs and parameters. In addition, the visualization results of each model on various datasets are shown in Figs. 7–10, with the deliberate exclusion of results from earlier purely convolution-based CD models (FC-EF, FC-SiamConc, and FC-Siam-Diff) for a better view. In particular, various colors are assigned to distinguish the correctness of the detection in different regions, including TP (white), TN (black), FP (red), and FN (green).

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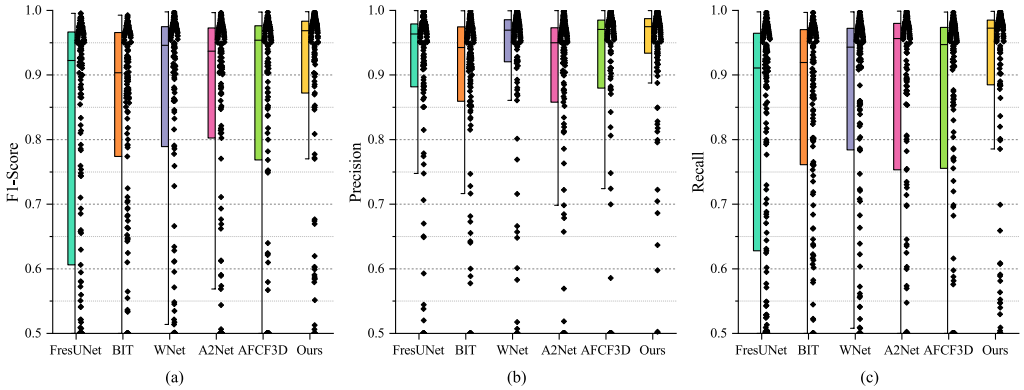


Fig. 11. (a) $F1$, (b) Pre, and (c) Rec values on test samples of the WHU dataset.

TABLE III

EVALUATION OF THE FINAL PROPOSAL WITH DIFFERENT VARIANTS ON FOUR DATASETS. THE BEST RESULTS ARE MARKED IN **BOLD**. THE RED TRIANGLE REPRESENTS THE LEAD PERCENTAGE

Network mode	CLCD		GZ		WHU		SYSU	
	F1(%)	IOU(%)	F1(%)	IOU(%)	F1(%)	IOU(%)	F1(%)	IOU(%)
Baseline	63.14▲15%	46.72▲22%	83.06▲6%	71.03▲10%	87.11▲5%	77.16▲9%	74.56▲10%	65.17▲6%
Baseline + SE	65.16▲11%	48.13▲19%	85.49▲3%	74.67▲5%	88.47▲3%	80.45▲5%	80.38▲2%	67.45▲4%
Baseline + TE	68.95▲6%	49.42▲16%	84.71▲4%	72.84▲7%	89.28▲2%	80.79▲4%	79.45▲4%	66.69▲5%
Baseline + SE +TE	**72.76**	**57.19**	**87.77**	**78.21**	**91.51**	**84.35**	**82.29**	**69.91**

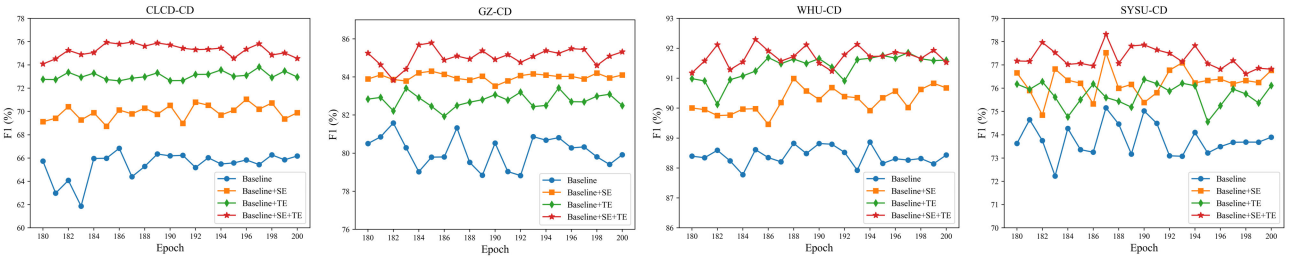


Fig. 12. Results of $F1$ for the validation set on four datasets.

Visualization on CLCD-CD (See Fig. 7): CLCD-CD encompasses a diverse array of change targets, including farmland, buildings, roads, and so on, for instance, villages in Fig. 7(a) and (d), croplands in Fig. 7(b) and (c), narrow roads in Fig. 7(e) and (g), and vegetation in Fig. 7(f) and (h). Despite the dual challenges of limited sample size and a broad spectrum of target types, our STENet consistently delivers outstanding visualization results, closely aligning with those presented in Table I. As shown in Fig. 7(d), where two houses emerge in the open space at the screen center, our model exhibits fewer missed regions compared to the other models, affirming the network's proficiency in detecting complex scenes. Similarly, in Fig. 7(f), amid strong seasonal and lighting variations, P2VNet and AFCF3D suffer substantial missed detection areas, while the remaining comparison models also perform poorly with extensive misdetection. In contrast, our model captures relatively clearer boundary information, yielding optimal visual performance.

Visualization on GZ-CD (See Fig. 8): Fig. 8 displays representative samples from the GZ-CD dataset, showcasing the urbanization process in the outskirts of Guangzhou, China. In Fig 8(a), small buildings at the edges are relatively easier to detect, resulting in slightly better performance for EGRCNN, ICIFNet, P2VNet, LCANet, and our STENet. However, other methods still encounter challenges related to missed detection. Notably, in buildings with complex boundary shapes in Fig. 8(d), our proposal excels in capturing clearer boundaries of the changing areas. When confronted with seasonal changes and light-shadow disturbances in Fig. 8(g), STENet enjoys significantly fewer missed and misdetected regions compared to comparison models. In summary, STENet achieves state-of-the-art performance for the qualitative results on GZ-CD, consistent with the quantitative results presented in Table I.

Visualization on WHU-CD (See Fig. 9): Similar to the sample selection in GZ-CD shown in Fig. 8, we chose several different samples from WHU-CD to facilitate the comparison of visualization performance, including a single building in Fig. 9(a) and (b) with different shooting angles, the more buildings in Fig. 9(c) and (d) located in the lower left corner, complex buildings in Fig. 9(e) obscured by vegetation, and disappearing buildings in Fig. 9(f)–(h). In Fig. 9(a) and (b), STENet successfully identifies the boundaries of the entire building more comprehensively. The challenge in Fig. 9(g) lies in the severe interference caused by the green roof building, resembling vegetation cover. Thanks to the semantic information between bitemporal images provided by spatial–temporal modeling, P2V and our STENet exhibit a superior ability to distinguish foreground from

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TABLE IV

ABLATION STUDIES. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY. (a) DIFFERENT NUMBERS OF TOP-k ATTENTION REGIONS. (b) SCHEMES FOR DSA. (c) EFFICIENCY ANALYSIS OF ATTENTION COMPUTATION. (d) SCHEMES FOR CONSTRUCTING PSEUDOVIDEO SERIES. (e) SCHEMES FOR FEATURE EXTRACTION IN TE. (f) HYPER-PARAMETER SELECTION FOR AUXILIARY LOSS

(a)

Top-k regions number	FLOPs	CLCD	GZ	WHU	SYSU
(2, 4, 8)	14.71	71.91	87.46	90.74	80.89
(4, 8, 16)	14.72	72.42	87.54	91.21	81.56
(4, 16, 32)	14.73	**72.76**	**87.77**	**91.51**	**82.29**
(8, 32, 48)	14.76	71.59	86.93	90.98	81.75

(b)

Attention	CLCD	GZ	WHU	SYSU
Self Attention	69.24▲5.1%	87.09▲0.7%	89.94▲1.7%	79.87▲3.0%
Cross Attention	69.49▲4.7%	86.75▲1.2%	90.41▲1.2%	80.52▲2.2%
DSA	**72.76**	**87.77**	**91.51**	**82.29**

(c)

Attention	FLOPs(G)	Inference speed	Throughput
Swin Transformer	15.21	0.382▼10.1%	2.65▲12.5%
Cross Attention	13.05	0.365▼5.6%	2.74▲8.8%
Quadtree Attention	16.20	0.406▼17.3%	2.42▲23.1%
DSA	**14.73**	**0.346**	**2.98**

(d)

Method	CLCD	GZ	WHU	SYSU
Linear Interpolation	72.54▲0.3%	87.24▲0.6%	91.46▲0.05%	82.06▲0.3%
Conv1 × 1	72.15▲0.8%	87.17▲0.7%	91.32▲0.21%	82.13▲0.2%
Ours	**72.76**	**87.77**	**91.51**	**82.29**

(e)

Method	FLOPs	CLCD	GZ	WHU	SYSU
3D-Transformer	20.16	**72.41**	**88.04**	**91.38**	**82.42**
3D-LSTM	18.24	72.19	87.38	91.34	81.92
2D-CNN	**11.79**	70.85	86.72	89.30	81.23
Ours(3D-CNN)	**14.73**	**72.76**	**87.77**	**91.51**	**82.29**

(f)

λ	CLCD	GZ	WHU	SYSU
0	70.64	86.23	89.58	80.69
0.2	71.95	**87.64**	90.91	**82.03**
0.5	**72.76**	**87.77**	**91.51**	**82.29**
0.8	**72.52**	87.14	**91.63**	81.75
1	72.27	86.76	91.32	81.26

background compared to others. Furthermore, in Fig. 9(h), under the lighting interference, while all other comparison models exhibit large missed areas, our STENet maintains higher positive rates and lower commission errors.

Visualization on SYSU-CD (See Fig. 10): Similar to the CLCD-CD dataset, SYSU-CD exhibits various types of variations, for instance, ships in Fig. 10(b) and (c), roads in Fig. 10(d) and (h), and buildings in Fig. 10(a), (e), (f), and (g). Specifically, in Fig. 10(e) and (f), STENet exhibits a noticeable lead in the detection of building. Furthermore, in Fig. 10(b) and (c), our proposal again demonstrates proficiency in detecting ships. Taking into account all four datasets, STENet consistently achieves the highest recognition scores with the smallest false positive and miss rates compared to the competitors. This highlights its suitability and superiority in the CD in RS images.

Taking WHU as an example, Fig. 11 further gives the statistical analysis of the $F1$/Pre/Rec values of the test pairs. In particular, the results of our model are distributed in a relatively high region compared to others. Overall, the qualitative comparisons and quantitative results confirm closely with each other, validating that our method has achieved a new state-of-the-art level.

C. Ablation Analysis

Intended to further validate the impact of various modules and branches on the practical performance, as shown in Tables III and IV, we carry out comprehensive ablation experiments in this section.

1) Design Choices for Different Network Modules: To underscore the importance of simultaneously incorporating the SE and TE through parallel branching, we present the ablation results in Table III, in which "baseline" is defined as the Siamese encoder and decoder without any new supplements. We primarily use $F1$ and IOU for the numerical comparison. Fig. 12 further displays the results on the validation set from Epoch 180 to 200. It is evident that the addition of either the "SE" module or the "TE" module enjoys better results in varying degrees on four datasets. With the two modules jointly considered, the results exhibit a more significant improvement, highlighting the complementary nature of the contributions achieved by using both encoders in parallel. Notably, on the CLCD-CD dataset, "Baseline + SE + TE" improves the $F1$ from 63.14% to 72.76% and the IOU by 22% compared to "Baseline." Fig. 12 exhibits the $F1$ curves for these methods in a more intuitive way. It can be seen that "Baseline + SE + TE" consistently outperforms the others, securing the highest performance. This illustrates that our proposed dual-branching treatment facilitates the extraction of more distinguishable change information from both spatial and temporal domains, thereby enhancing the overall characterization of the model.

2) Impact of Top-k Attention Region Number: To optimize training efficiency and prevent potential performance degradation, it is advisable to consider chunking S as a divisor of the training size, eliminating the need for padding operation. As an illustration, for an image with a resolution of 256 (8×32), we set $S = 8$, ensuring that it serves as a divisor for the feature map at each stage, resulting in a total of 64 regions. To accommodate decreasing region sizes in later stages, the value of top-k is gradually increased, striking a balance for maintaining a reasonable number of tokens. Our ablation results in Table IV(a) elucidate the impact of varying top-k regions on the three-stage spatial encoder. As the top-k value incrementally increases along with different phases, a performance improvement can be easily observed, which is attributed to the participation of a greater number of tokens in the attention computation. Notably, the highest $F1$ is achieved when the top-k values are sequentially set at 4, 16, and 32. However, a slight decline in performance is discerned when the top-k settings are 8, 32, and 48. We surmise that an excessive number of participating tokens may disrupt the

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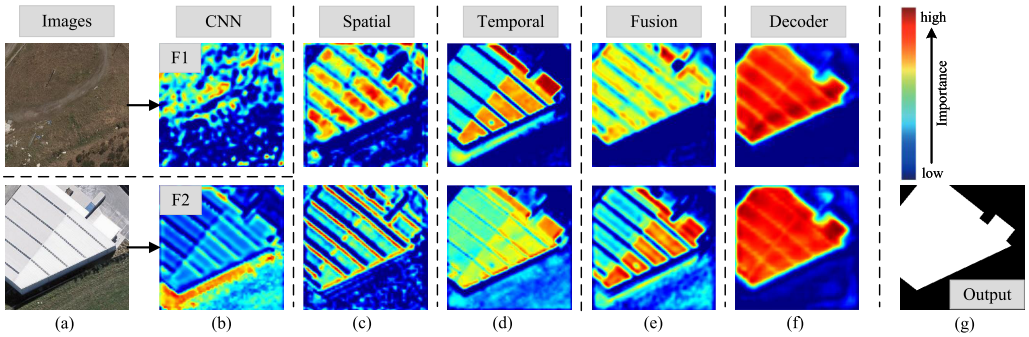


Fig. 13. Network visualization taking images from WHU-CD as examples. (a) Input images. (b) Selected feature maps F_1 and F_2 generated by CNN. (c) and (d) Selected feature maps obtained by SE and TE. (e) Selected feature maps coupled through the STF module. (f) Change probability maps from the final layer of the decoder. (g) Change map.

explicit sparsity constraint in CD tasks since the areas of actual change are sparse, which constitute only a small fraction of the entire image.

3) Improvements in Attention Schemes: The traditional dot-product SelfAtt mechanism calculates only the similarity between the representation vectors of a single RS image, while the CrossAtt mechanism explores semantic connections between the bitemporal images but leads to large computational redundancy. In contrast, during the design of STENet, we introduce a DSA to achieve functional interaction by dynamically establishing semantic similarity connections between bitemporal RS images. As shown in Table IV(b), one can easily observe that the model using DSA consistently outperforms the $F1$ values in all cases, demonstrating the superiority of generalization.

4) Efficiency Analysis of Attention Computation: To demonstrate the computational efficiency of the DSA mechanism, we conduct a comparative analysis of inference speed and throughput across models employing different attention schemes. Specifically, inference speed is quantified by measuring the time taken to complete one inference iteration on a single sample. The throughput is defined as the sample numbers processed per second. By utilizing a uniform set of validation samples from the GZ dataset, we determine throughput values by dividing the total number of validation samples by the total time elapsed, thus yielding the per-second running efficiency. For fairness, we make the network structure of all comparison models consistent with STENet, with only different attention modules replaced. As detailed in Table IV(c), the results by utilizing Swin transformer, CrossAtt, and quadtree attention mechanisms are given under float32 Pre. It is observed that CrossAtt and Swin transformers enjoy appealing inference speeds and throughput, attributed to their structural simplicity. Comparatively, quadtree attention delivers diminished running efficiency, a consequence of its recursive design that hinders computational parallelism. Finally, our DSA leverages an efficient approach in the collection of key–value tokens and employs the sparse matrix multiplication, culminating in the achievement of the highest throughput and the fastest inference rates.

5) Schemes for Constructing Pseudovideo Series: To examine the impact of different approaches to construct pseudovideo frames on model optimization, we compare the mix-up practice with a linear interpolation method (utilized in P2V) and a method based on a convolutional learning strategy (using conv1 $\times$ 1 in temporal dimension) in Table IV(d) while keeping the other model components unchanged. As observed in the table, the used construction technique outperforms both the linear interpolation and the more complex CNN-based learning strategy across all metrics.

6) Schemes for Feature Extraction in Temporal Encoder: To assess the impact of 3-D-CNN used in TE, the experiments replacing 3-D-CNN with other modules, e.g., Swin transformer and 3-D-LSTM, are conducted. These replacements are known for their long-range context-learning capability. We also compare 3-D-CNN with the 2-D-CNN counterpart. Note that it is required to flatten the pseudovideo by dimensionality reduction before extracting features using the 2-D-CNN module. According to the findings in Table IV(e), yet, substituting 3-D-CNNs with a Swin transformer would tap the potential of performance improvement slightly; the concurrent increasing of computational costs occurs. This outcome suggests that DSA may already provide adequate contexts in the TE. As for the 2-D-CNN counterpart, while less computational burden is needed, a notable degradation of model performance can also be seen. This decline is likely attributed to the loss of critical temporal information following dimensionality reduction in the pseudovideos. In conclusion, these results evidence the well balance of the chosen 3-D-CNN module, with attractive efficacy yet without excessively increasing computational load.

7) Hyperparameter Selection for Auxiliary Loss: In order to validate the effect of deep supervision on the temporal encoder, different settings of the hyperparameter λ are ablated. The experimental results are shown in Table IV(f), where we iterate through the parameter values in the range of {0, 0.2, 0.5, 0.8, 1} to report the performance variations. From this table, we can see that the imposed deep regulation definitely favors better performance. Besides, the results from different settings are relatively stable. Empirically, we consider the balance weight 0.5 as an appropriate choice.

D. Network Visualization

Aiming at visualizing the feature maps at successive stages of our STENet, we deliver an instance with images from WHU-CD. Through channelwise visualization, we select representative cases to be presented in Fig. 13. Given a pair of bitemporal inputs [see Fig. 13(a)], the multilevel feature

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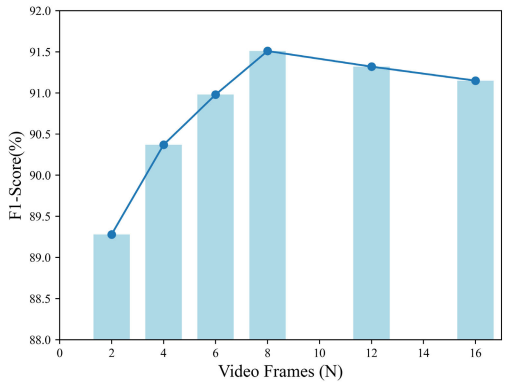


Fig. 14. Effect of the sequence length N.

maps F_1 and F_2 [see Fig. 13(b)], with rough localization of both targets and contours, are first captured by the Siamese CNN (ResNet18). Then, the information interaction between the dual-time F_1 and F_2 is performed by the spatial encoder module to generate the coupled feature map [see Fig. 13(c)]. It is obvious that thanks to our dynamic selection of attention mechanism, by combining sparse region selection and cross-attention, the interaction process directs the model's attention to the changing regions and enforces the suppression of task-irrelevant distractions. After that, we start from a sequence of time frames and obtain a temporal information-enriched feature map through a TE, which can be seen in [see Fig. 13(d)] showing more fine-grained boundary and texture features. In addition, the spatial and temporal feature fusion module is introduced to gradually aggregate the boundary details and global structures to obtain the advanced fusion features [see Fig. 13(d)]. Finally, we gradually upsample the feature map to the original size by a simple yet effective decoder module, as shown in Fig. 13(f). It can be seen that the attention is highly concentrated on the change region, which proves the overall effectiveness of our proposed modeling components.

E. Effect of the Sequence Length

By varying N in synthetic videos, the performance of our model is also ablated. Practically, we conduct six experiments on the WHU dataset, each with a different value of N. The results are presented in Fig. 14. It can be intuitively observed that prior to $N = 8$, F_1 gradually increases. As expected, given a relatively larger N, the model would extract more temporal information, thus enhancing the performance. However, with $N > 8$, more data redundancy happens, which leads to an overfitting issue and delivers a declined performance.

V. Conclusion

In this article, we propose an STENet for CD in RS images. The overall framework incorporates parallel encoding of spatial transformers and temporal CNNs. The former dynamically employs a sparse selection of interactive attention to locate the top-k semantically relevant regions between dual-temporal images for calculating attention weights, steering attention to areas that are truly changing. The latter employs suitable data augmentation techniques to construct temporal evolution sequences for dual-temporal images, followed by 3-D-convolution blocks that acquire additional temporal information. Experimental results demonstrate the robustness and generalizability of our proposed method, yielding optimal results across four popular CD datasets. Currently, the parameter quantity of our model is reasonable when juxtaposed with other benchmark models, yet the model is not lightweight enough to be deployed on mobile devices. Identifying model lightweighting is then a promising area for further study. In addition, we observe that the efficiency of our temporal branch falls short compared to models relying solely on 2-D convolution networks. A critical avenue for future enhancements involves addressing the shortfall in computational complexity. Moreover, explorations into methods for rendering objects at varying resolutions, including neural operators [38] and implicit representation functions [39], merit thorough investigation given that the region of change often differs across inputs.

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